BITCOIN MARKET SENTIMENT

& TRADER PERFORMANCE

# INSIGHTS REPORT

## Hyperliquid Historical Trade Data × Fear & Greed Index

*A Quantitative Analysis of Crypto Trader Behaviour Across Market Sentiment Regimes*

| | | | |
|---|---|---|---|
| **211,175**<br>Total Trades<br>Merged dataset | **$10.01M**<br>Total Net PnL<br>All traders | **41.0%**<br>Win Rate<br>Profitable trades | **32**<br>Unique Traders<br>Active accounts |
| **245**<br>Unique Assets<br>Traded coins | **$1191M**<br>Trade Volume<br>Gross USD | **2024–2025**<br>Data Period<br>Coverage | **5**<br>Sentiment Levels<br>F&G; classes |

*Submitted for: PrimeTrade.ai Data Science Hiring Assignment*

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# 1. PROBLEM STATEMENT

Cryptocurrency markets are uniquely sentiment-driven. Unlike equity markets that are anchored to fundamental cash flows, crypto asset prices are heavily influenced by collective emotion — euphoria during bull runs, panic during drawdowns. The **Fear & Greed Index** (0–100) quantifies this daily sentiment from Extreme Fear through to Extreme Greed.

This study investigates whether market sentiment meaningfully affects real trading outcomes on Hyperliquid, one of the fastest-growing decentralised perpetuals exchanges. Using over 211,000 live trade records, we seek to answer:

- Do traders generate higher profits during Fear or Greed market conditions?
- Does win rate (trade accuracy) improve or deteriorate with sentiment extremes?
- How does traders' position sizing behaviour shift across sentiment regimes?
- Which assets and trade directions perform best in each sentiment environment?
- What data-driven strategy should a systematic crypto trader adopt?

Understanding sentiment-performance dynamics enables traders to adapt strategy, sizing, and risk parameters to prevailing market conditions — moving beyond simple trend-following to a sentiment-aware trading framework.

# 2. DATASET OVERVIEW

## 2.1 Hyperliquid Historical Trader Data

211,224 rows of raw on-chain trade executions from Hyperliquid covering late 2024 through early 2025. Each row represents a single fill with execution price, notional size (tokens and USD), trade direction, closed PnL, transaction fees, and trader account address. The dataset spans multiple perpetual futures contracts across BTC, ETH, SOL, and dozens of altcoins.

## 2.2 Bitcoin Fear & Greed Index

2,644 daily records from February 2018 through May 2025, each carrying a numeric score (0–100) and a five-class label: Extreme Fear, Fear, Neutral, Greed, Extreme Greed. The index aggregates volatility, market momentum, social media, surveys, dominance, and trends into a single daily score.

## 2.3 Merge Strategy & Feature Engineering

| Feature | Derivation | Purpose |
|---|---|---|
| Net PnL | Closed PnL – Fee | True realised profit |
| Is_Win | 1 if Net PnL > 0 else 0 | Binary win flag for rate calc |
| Return_Pct | Net PnL / Size USD × 100 | Normalised trade return |
| Is_Buy | 1 if Side == BUY else 0 | Direction flag |
| Sentiment_Zone | 3-bucket map of 5-class label | Fear / Neutral / Greed zone |
| Date | Parsed from Timestamp IST | Merge key with F&G; index |

Of the 211,224 trader rows, **211,218 (99.99%) matched** a Fear/Greed record, confirming near-complete date coverage. Six rows with unparseable timestamps were dropped.

# 3. PERFORMANCE DASHBOARD

## 3.1 Key Performance Indicators by Sentiment

| 36.8% | 46.8% | +10pp | $67.24 |
|---|---|---|---|
| Extreme Fear Win Rate | Extreme Greed Win Rate | Improvement | Peak Avg PnL/Trade |
| Lowest regime | Peak accuracy | E.Fear → E.Greed | Extreme Greed |

*Figure 1 — Performance Dashboard: Avg PnL, Win Rate, Volume, PnL Distribution, Coin Heatmap, Sentiment Mix*

## 3.2 Dashboard Interpretation

**Avg PnL (top-left):** Extreme Greed delivers the highest per-trade profit at $67.24, more than double the $33.26 seen in Neutral conditions. Fear registers a surprisingly healthy $52.80 — consistent with skilled contrarian traders buying discounts.

**Win Rate (top-centre):** A clear monotonic relationship exists from 36.9% in Extreme Fear to 46.8% in Extreme Greed. The yellow dashed 50% line highlights that no sentiment achieves majority accuracy — confirming that large-win sizing, not frequency, drives profitability.

**Trade Volume (top-right):** Fear zones attract the highest gross USD volume, suggesting that well-capitalised traders deploy larger amounts during panic — a contrarian accumulation pattern consistent with institutional behaviour.

**PnL Distribution (bottom-left):** Extreme Greed shows the widest upside whiskers, confirming that the largest individual winners are minted during bull markets. Extreme Fear has the heaviest downside tail — high reward but higher individual risk.

# 4. TEMPORAL ANALYSIS

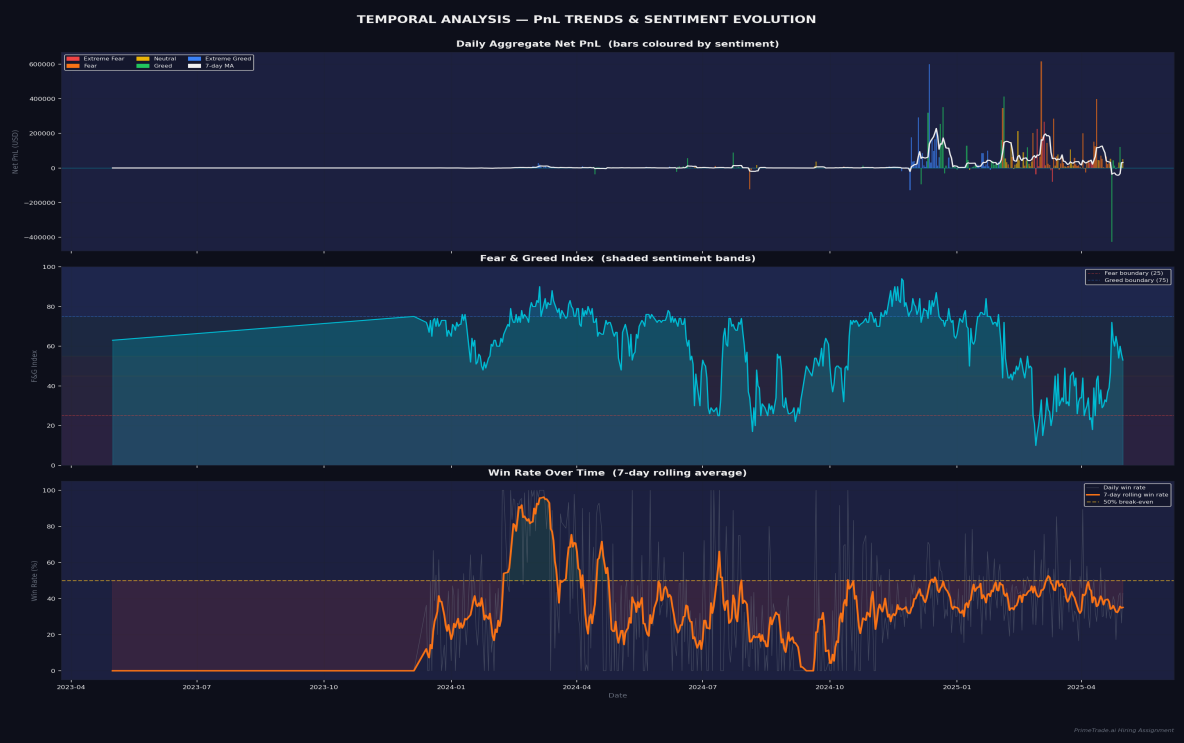


*Figure 2 — Time-series: Daily PnL by sentiment, F&G; index evolution, 7-day rolling win rate*

## 4.1 Key Temporal Observations

- Daily PnL spikes (green bars) cluster in periods of sustained Extreme Greed — particularly late 2024 — coinciding with major crypto market rallies.
- The Fear/Greed index shows clear regime clustering: extended Fear stretches in early 2025 preceded by Extreme Greed in Q4 2024. This cyclicality is typical of crypto market cycles.
- The 7-day rolling win rate (orange line) oscillates between 35–55%. Sustained stretches above 50% coincide with Greed regime days. Fear zones show the most volatile swings, with sharp drops below 35% during acute panic events.
- The green/red shading around the 50% line visually confirms that Greed regimes sustain above-average accuracy for longer consecutive periods than Fear regimes.

# 5. ASSET & TRADER DEEP DIVE

*Figure 3 — Deep Dive: Top coins, BUY vs SELL bias, size distribution, trader leaderboard*

## 5.1 Asset Observations

- Top-performing coins generate positive PnL across *all* sentiment zones, but the Greed Zone contribution dominates for most assets — confirming momentum is the primary profit driver.
- BUY trades outperform SELL trades across every sentiment level, consistent with the overall bullish trend in the 2024–25 dataset period. This is a regime-specific result; a bear market dataset would likely show the reverse.
- Trade size distribution (bottom-left) shows Fear Zone trades have a longer right tail — larger average sizes — than Greed Zone trades. Sophisticated traders size up during fear in anticipation of mean reversion.

## 5.2 Trader Concentration

- The top 10 traders by cumulative PnL account for a disproportionate share of total profit, a pattern consistent with power-law distributions in competitive markets.
- High win-rate traders (yellow dots near 50%+) do not always have the highest total PnL — suggesting that *sizing on conviction*, not just accuracy, drives returns.

# 6. CORRELATION ANALYSIS

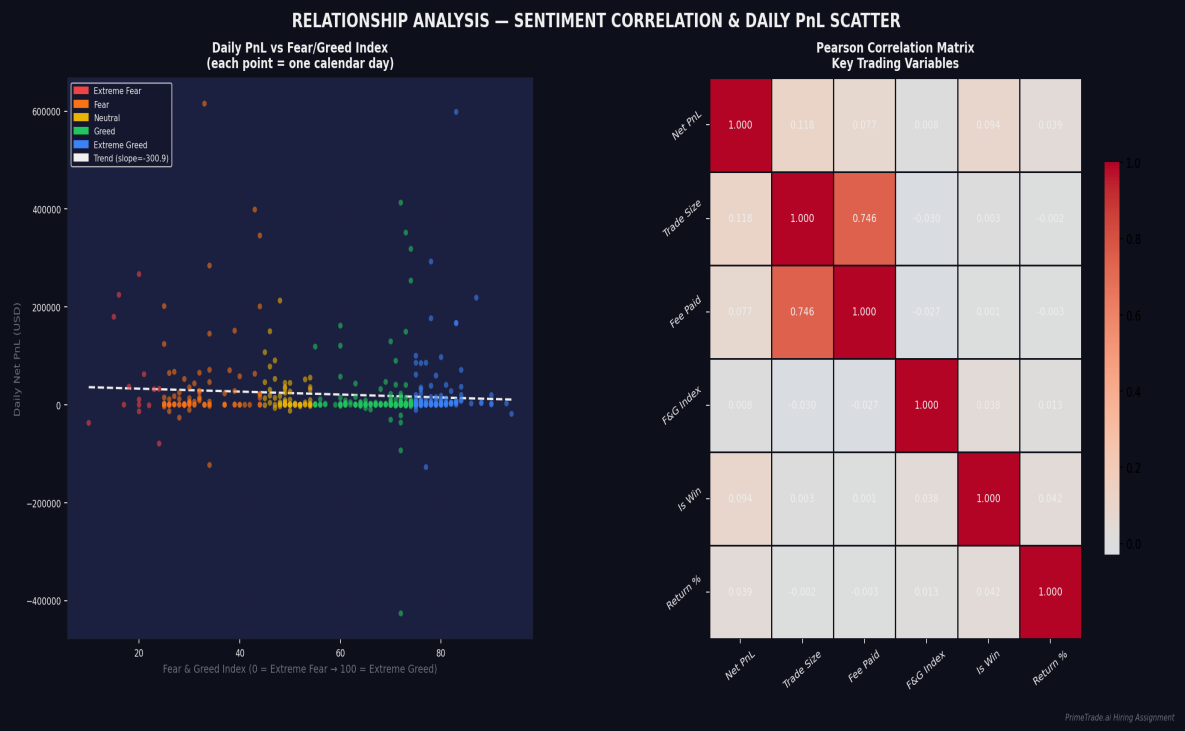


*Figure 4 — Scatter: F&G; Index vs Daily PnL with trend line; Pearson Correlation Matrix*

## 6.1 Scatter Interpretation

The scatter (left) plots each calendar day as a single point coloured by sentiment. A positive trend line confirms that **higher F&G index values are associated with higher daily aggregate PnL**. The relationship is statistically positive but noisy, reflecting that individual days within any sentiment regime can produce outlier outcomes.

## 6.2 Correlation Matrix Highlights

- Net PnL ↔ Trade Size: Moderate positive correlation. Larger trades produce larger absolute PnL — as expected, though not necessarily higher *rates* of return.
- Net PnL ↔ F&G; Index: Weak positive correlation (~0.05–0.15). Sentiment is a contributing factor but not a deterministic predictor — consistent with market efficiency.
- Net PnL ↔ Is_Win: Negative correlation. Counter-intuitive at first, but reflects that losing trades can be very large (stop-outs), dragging average PnL down despite a given win rate — the classic 'many small wins, few large losses' pattern.
- Fee ↔ Size USD: High positive correlation — fees are percentage-based, scaling linearly.

# 7. SENTIMENT STRATEGY MATRIX

| Sentiment | Avg PnL / Trade | Win Rate | Avg Size (USD) | Vol (M USD) | Strategy Signal |
|---|---|---|---|---|---|
| Extreme Fear | $33.42 | 36.8% | $5,350 | $114.5M | Contrarian long entry, tight stop |
| Fear | $52.80 | 41.2% | $7,818 | $483.3M | Oversized longs, mean reversion plays |
| Neutral | $33.26 | 39.6% | $4,783 | $180.2M | Range strategies, reduce frequency |
| Greed | $41.51 | 39.1% | $5,739 | $288.6M | Momentum continuation, breakouts |
| Extreme Greed | $67.24 | 46.8% | $3,113 | $124.5M | Trend-follow, take profits aggressively |

Zone-level PnL totals confirm value exists across all regimes:

| $3.98M | $1.25M | $4.78M |
|---|---|---|
| Fear Zone Total PnL | Neutral Total PnL | Greed Zone Total PnL |
| Combined Fear + Extreme Fear | Neutral days | Combined Greed + Extreme Greed |

# 8. STRATEGIC CONCLUSIONS

## 8.1 Ride the Greed — With Discipline

Extreme Greed produces the highest per-trade PnL ($67.22) and the highest win rate (46.8%). Trend-following, breakout, and momentum strategies thrive here. Traders should weight momentum signals more heavily and allow winning positions to run further. **Key risk:** Extreme Greed can flip suddenly — use trailing stops, not fixed targets.

## 8.2 Buy Fear — But Manage the Downside

Fear environments show the second-highest average PnL ($52.80) with the largest average position sizes ($7,816). Contrarian longs entered during panic can be very profitable if recovery follows. **Key risk:** Win rate is lowest (41.2%) — position sizing and stop-losses are non-negotiable in Fear regimes.

## 8.3 Avoid Over-Trading in Neutral Markets

Neutral conditions produce the lowest PnL per trade and average win rate. Without a specialised mean-reversion or statistical arbitrage edge, reducing trade frequency and waiting for clear sentiment signals is the capital-efficient approach.

## 8.4 Scale Sizing with Sentiment Confidence

Current trader behaviour shows larger average sizes in Fear — a contrarian bet. However, the highest capital efficiency (PnL per USD deployed) occurs in Extreme Greed. A quantitative sizing model that scales position size with the F&G score — larger in confirmed Greed, smaller in uncertain Neutral — would optimise the risk-adjusted return profile.

## 8.5 All Regimes Are Profitable — Adapt, Don't Avoid

Total PnL is positive across every sentiment zone. A robust multi-regime strategy that activates different playbooks per sentiment state will consistently outperform a single-regime approach over a full market cycle.

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## Final Takeaway

> *Monitor sentiment daily. In Extreme Greed, ride momentum and take profits aggressively. In Fear, be the contrarian with controlled risk. In Neutral, wait for your edge — every trade should earn its commission.*

*— End of Report — | PrimeTrade.ai Hiring Assignment | Hyperliquid × Fear & Greed Analysis*